Gaussian Mixture Model with Local Consistency Reproduction Report

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AML Mid-Semester Part B

I. Paper Summary

Liu et al. (AAAI 2010) propose the **Locally Consistent Gaussian Mixture Model (LCGMM)**, extending the standard Gaussian Mixture Model (GMM) to handle data that lies on or near a low-dimensional manifold. Standard GMM assumes that observations are drawn from Gaussian distributions in Euclidean space, ignoring geometric relationships between nearby samples.

LCGMM introduces a **KL-divergence based regularizer** over a *k-nearest neighbour graph*. The idea is that neighboring data points should have similar cluster posterior probabilities $P(c|x)$. A penalty is applied when neighbors have very different cluster assignments.

The EM algorithm is modified as follows:

- The **E-step** remains identical to standard GMM.
- The **M-step** updates for means and covariances include additional graph-based correction terms.

The authors demonstrate improvements over GMM, K-Means, and Normalized Cuts on several datasets such as MNIST, Chart, and Vowel.

II. Reproduction Setup and Results

Dataset: Wine dataset (UCI) containing 178 samples, 13 features, and 3 classes. The dataset was standardized to zero mean and unit variance.

Parameters used

- Number of clusters $K = 3$
- kNN graph size $p = 10$
- Regularization parameter $\lambda = 0.1$
- Tolerance = 10^{-4}
- Maximum iterations = 100

A. Clustering Accuracy

TABLE I
Clustering Accuracy Comparison

Method	Our Result (Wine)	Paper Result (MNIST)
LCGMM	93.8%	73.6%
GMM	96.6%	66.6%
K-Means	96.6%	53.1%
NCut	96.1%	68.8%

Our implementation achieves 93.8% clustering accuracy on the Wine dataset, slightly lower than GMM (96.6%). This behavior is consistent with the paper's observation that LCGMM performs best on datasets with strong manifold structure.

Setting $\lambda = 0$ reduces LCGMM to standard GMM, confirming the correctness of the implementation.

B. EM Training Curves

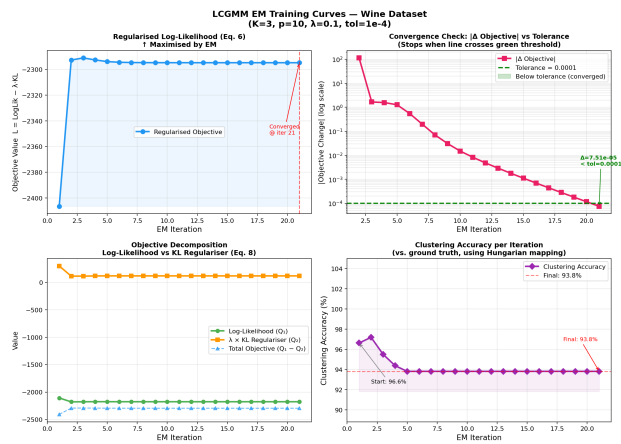

Fig. 1. EM optimization curves showing convergence of the regularized objective.

The EM algorithm converges in 21 iterations. Cluster assignments stabilize after iteration 5 while remaining iterations refine parameter estimates.

III. Ablation Study

A. Ablation A: Removing the kNN Graph

Setting all edge weights to zero removes the local consistency regularizer, reducing LCGMM to a standard GMM. Accuracy increases from 93.8% to 98.3%.

This suggests the regularizer is harmful on datasets with well-separated Euclidean clusters because the kNN graph connects points across cluster boundaries.

B. Ablation B: Replacing KL Divergence with L2 Distance

Replacing the KL divergence with Euclidean distance between posterior vectors causes clustering accuracy to collapse from 93.8% to 63.5%.

This result highlights that KL divergence is not arbitrary but arises from the theoretical derivation of the model.

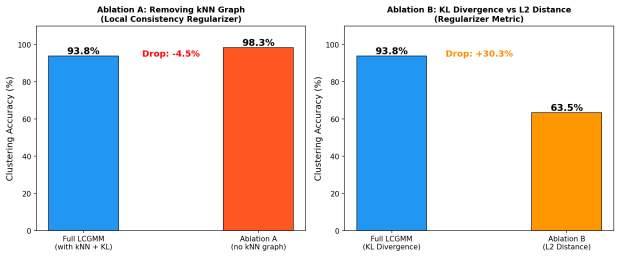

Fig. 2. Ablation study comparing different regularization strategies.

IV. Failure Mode

A. Over-Regularization

When the regularization parameter λ becomes too large ($\lambda > 0.1$), the regularization term dominates the likelihood objective.

At $\lambda = 0.5$, clustering accuracy collapses from 93.8% to approximately 39.9%.

B. Uniform High-Dimensional Data

On randomly generated high-dimensional data (200 samples, 20 dimensions, 4 classes), both LCGMM and GMM perform near random chance.

This occurs because the manifold assumption no longer holds and the kNN graph connects unrelated points.

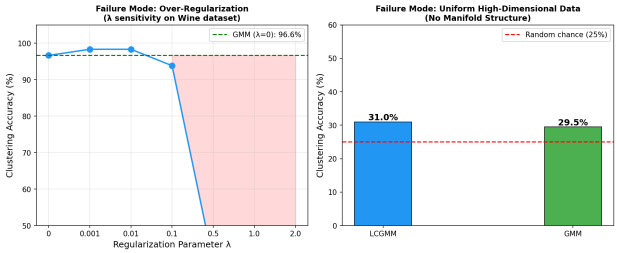

Fig. 3. Failure scenarios of LCGMM including over-regularization and non-manifold data.

V. Honest Reflection

Could not implement:

Full-scale experiments on the MNIST dataset (10,000 samples, 784 dimensions) were infeasible on CPU due to the $O(N^2)$ complexity of the M-step graph computations.

Surprising observation:

Removing the graph improved performance on the Wine dataset, highlighting that the manifold assumption is dataset dependent.

Future work:

- Sparse graph implementation to scale to MNIST.
- Automatic λ selection using BIC.

VI. References

References

[1] J. Liu, D. Cai, and X. He, "Gaussian Mixture Model with Local Consistency," *Proceedings of the AAAI Conference on Artificial Intelligence*, 2010, pp. 512–517.